

ARMA(1,2) or ARMA(1,2) 1 -

- 1- $H_0: \ell_K = 0$ for $K=1, 2, \dots, m$
- 2- Level of significance. $\alpha = 0.05 = 5\%$
- 3- Test statistic: Ljung Box Pierce test

$$Q_m^* = n(n+2) \sum_{K=1}^m \frac{\ell_K^2}{n-K}, \text{ with } m=5, n=50$$

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K :	1	2	3	4	5	6	7	8	9	10
ℓ_K :	-0.15	0.08	-0.05	0.01	0.03	0.01	0.05	-0.04	0.01	0.02

$$Q_5^* = 50(50+2) \sum_{K=1}^5 \frac{\ell_K^2}{50-K}$$

$$= (50)(52) \left[\frac{(-0.15)^2}{50-1} + \frac{(0.08)^2}{50-2} + \frac{(-0.05)^2}{50-3} + \frac{(0.01)^2}{50-4} + \frac{(0.03)^2}{50-5} \right]$$

$$= 1.74$$

5 Critical region:

$$\chi^2 \geq \chi_{\alpha, m-p-q}^2$$

$$\chi^2 \geq \chi_{(0.05, 2)}^2 = 5.99$$

$$\chi^2 \geq 5.99$$

$$m-p-q = 5-1-2 = 2$$

$$\therefore m > p+q$$

$$5 > 1+2$$

$$5 > 3$$

Since $1.74 < 5.99$, we do not reject the null

hypothesis. So we recommend that fitted model $ARM(1, 2)$ is an adequate model.



Given AR(1) process: $(1 - 0.7B)(X_t - 50) = Z_t$

$$X_t - 0.7BX_t - 50 + (0.7)(50)B = Z_t$$

$$X_t - 0.7X_{t-1} - 50 + (50)(0.7) = Z_t$$

$$X_t = 0.7X_{t-1} - (50)(0.7) + Z_t + 50$$

$$X_t = 0.7(X_{t-1} - 50) + (Z_t + 50)$$

put $t = n+L$

$$X_{n+L} = 0.7(X_{n+L-1} - 50) + Z_{n+L} + 50$$

Take Conditional expectation

$$X_n(l) = 0.7 (X_n(l-1) - 50) + z_n(l) + 50$$
$$l=1; X_n(1) = 0.7 (X_n - 50) + z_n(1) + 50$$

$$\therefore X_n(0) \rightarrow X_n; z_n(1) = 0$$

$$\boxed{X_n(1) = 50 + 0.7 (X_n - 50)}$$

$$l=2; X_n(2) = 0.7 (X_n(1) - 50) + z_n(2) + 50$$

$$X_n(2) = 50 + 0.7 (X_n(1) - 50)$$

$$X_n(2) = 50 + 0.7 [50 + 0.7 (X_n - 50) - 50]$$

$$X_n(2) = 50 + (0.7)^2 (X_n - 50)$$

$$\boxed{X_n(l) = 50 + (0.7)^l (X_n - 50)}$$

put $n=100$ in above equation

$$X_{100}(l) = 50 + (0.7)^l (X_{100} - 50)$$

$$X_{100}(1) = 50 + (0.7) (X_{100} - 50)$$

$$X_{100}(1) = 50 + (0.7) (52 - 50)$$

$$\boxed{X_{100}(1) = 51.4}$$

Since

$$X_{99} = 49$$

$$X_{100} = 52$$

$$X_{100}(2) = 50 + (0.7)^2 (X_{100} - 50)$$

$$= 50 + (0.7)^2 [52 - 50]$$

$$X_{100}(2) = 50.98$$

(b) obtain 95% forecast limits for X_{101} , if $\sigma_z^2 = 5$

$$X_n(l) \pm Z_{\alpha/2} \sqrt{\sigma_z^2 \sum_{i=0}^{l-1} \psi_i^2}$$

$$X_n(l) \pm 1.96 \sqrt{\sigma_z^2 \sum_{i=0}^{l-1} \psi_i^2}$$

$$X_{100}(1) \pm 1.96 \sqrt{(5) \psi_0^2}$$

$$51.4 \pm 1.96 \sqrt{5 \times 1}$$

$$51.4 \pm 4.38$$

$$47.02 < X_{101} < 55.78$$

$$(1-\alpha)\% = 95\%$$

$$1-\alpha = 0.95$$

$$\alpha = 1 - 0.95$$

$$\alpha = 0.05$$

$$Z_{\alpha/2} = Z_{0.05/2}$$

$$= Z_{0.025}$$

$$= 1.96$$

$$\psi_0 = 1$$

①

$Y_t - 53 = 0.65(Y_{t-1} - 53) + Z_t$
is fitted to an observed time series of 100 observations. Forecast Y_{101} and Y_{102} .

(i) Compute the MMSE forecast of Y_{101} and Y_{102} if $Y_{99} = 52.7$ and $Y_{100} = 53.5$ using
a) $\psi(B)$ polynomial b) $\hat{\pi}(B)$ polynomial

(ii) Suppose that Y_{101} comes out be 51.3. update the forecast of Y_{102} by shifting forecast origin at $t = 101$.

$$Y_t = 53 + 0.65(Y_{t-1} - 53) + Z_t$$

$$t = n+l, \quad Y_{n+l} = 53 + 0.65(Y_{n+l-1} - 53) + Z_{n+l}$$

Taking the conditional expectation

$$Y_n(l) = 53 + 0.65(Y_{n(l-1)} - 53) + Z_n(l)$$

$$l=1; \quad Y_n(1) = 53 + 0.65(Y_n - 53) + Z_n(1)$$

$$Y_n(1) = 53 + 0.65(Y_n - 53)$$

$$l=2 \quad Y_n(2) = 53 + 0.65(Y_n(1) - 53)$$

$$= 53 + 0.65[53 + 0.65(Y_n - 53) - 53]$$

$$Y_n(2) = 53 + 0.65^2(Y_n - 53)$$

(2)

$$Y_n(l) = 53 + (0.65)^l [Y_n - 53]$$

$$\begin{aligned} n=100 \quad Y_{100}(l) &= 53 + (0.65)^l [Y_{100} - 53] \\ &= 53 + (0.65)^l [53.5 - 53] \end{aligned}$$

$$Y_{100}(l) = 53 + 0.5 (0.65)^l$$

$$\psi_l = \phi^l (0.65)^l$$

$$\begin{aligned} l=1; \quad Y_{100}(1) &= 53 + 0.5 (0.65)^1 \\ Y_{100}(1) &= 53.325 \end{aligned}$$

$$\begin{aligned} l=2; \quad Y_{100}(2) &= 53 + 0.5 (0.65)^2 \\ Y_{100}(2) &= 53.21 \end{aligned}$$

$$(ii) \quad \hat{y}_{n+1}(l) = \hat{y}_n(l+1) + \psi_l [\hat{y}_{n+1} - \hat{y}_n(1)]$$

$$\begin{aligned} n=100 \quad \hat{y}_{101}(l) &= \hat{y}_{100}(l+1) + \psi_l [\hat{y}_{101} - \hat{y}_{100}(1)] \\ &= \hat{y}_{100}(l+1) + \psi_l [51.3 - 53.325] \end{aligned}$$

$$\hat{y}_{101}(l) = \hat{y}_{100}(l+1) - 2.025 \psi_l$$

$$\begin{aligned} l=0; \quad \hat{y}_{101}(0) &= \hat{y}_{100}(1) - 2.025 \psi_0 \\ &= 53.325 - 2.025(1) = 51.3 \end{aligned}$$

$$\begin{aligned} l=1; \quad \hat{y}_{101}(1) &= \hat{y}_{100}(2) - 2.025 \psi_1 \\ &= 53.21 - 2.025(0.65) \\ \hat{y}_{101}(1) &= 53.21 - 1.32 \Rightarrow 51.89 \end{aligned}$$